Weighted Adaptive Stimulation After Batch-testing Identification (WASABI) Problem Formulation

Problem Formulation

There are a few ways we can formulate the question

1. Similar to Draelos 2020 first identify the binary connections using group testing, then use the identified connections to fit a weighted model.
2. Directly use a linear-Poisson model to identify weights using group testing. (Works on the assumption that connections are linear)
3. If not linear, use a neural network or some other nonlinear approach.

We will focus on (2) first, with fixed continuous stimulation parameters (e.g., current, frequency...)

Goal

To develop a method for adaptive stimulation that iteratively identifies the most informative stimulation sites. The dynamics of the reach is modeled using a linear-Poisson model , using a weighted linear-Poisson model to identify the most informative stimulation sites and improve the accuracy of the model.

Dynamics model

We model stimulation as a perturbation to a latent dynamical system through the mapping $\boldsymbol{B}$. Specifically, we model the system using a Poisson linear dynamical system (PLDS), which combines firing rate estimation with dimensionality reduction. We assume linear, Markovian (AR(1)) state evolution, Poisson emissions, and stationary $\boldsymbol{A}$. Eventually, we can have a Kalman-like update to allow for nonstationarity in $\boldsymbol{C}$ and a compositional model for $\boldsymbol{A}$.

State space equation

$$\begin{aligned} \boldsymbol{x}_0 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), & \boldsymbol{u} &= \text{softplus}\big(\boldsymbol{V}\boldsymbol{s}_{t_{stim}} + \boldsymbol{b}\big) \\ \boldsymbol{x}_t &= \boldsymbol{A}\boldsymbol{x}_{t-1} + \boldsymbol{B}\boldsymbol{u} + \boldsymbol{\epsilon}_t, & \boldsymbol{\epsilon}_t &\sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \\ \boldsymbol{y}_t &\sim \text{Poisson}\big(\text{softplus}(\boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{d})\big) \end{aligned}$$

Variables

- $\boldsymbol{x}_t \in \mathbb{R}^K$: latent state at time t
  - K is the latent dimensionality
- $\boldsymbol{y}_t \in \mathbb{Z}_{\geq 0}^N$: binned spike count vector at time t
  - N is the number of channels
- $\boldsymbol{s}_t \in \{0,1\}^N$: one-hot stim indicator at time t
  - zero vector on control trials
  - stimulation occurs at $t = t_{\text{stim}}$

Hyperparameters

- $K \in \mathbb{Z}_+$: latent dimensionality
  - $K \ll N$, low-rank assumption
- $w \in \mathbb{R}_+$: stim spatial spread radius
- $\boldsymbol{V} \in \mathbb{R}^{N \times N}$: Toeplitz matrix encoding spatial current spread

Fitted parameters

- $\boldsymbol{A} \in \mathbb{R}^{K \times K}$: dynamics operator
- $\boldsymbol{B} \in \mathbb{R}^{K \times N}$: stimulus-to-latent mapping
- $\boldsymbol{b} \in \mathbb{R}^K$: perturbation threshold
  - what stim strength would perturb latents
- $\boldsymbol{C} \in \mathbb{R}^{N \times K}$: observation (loading) matrix
- $\boldsymbol{d} \in \mathbb{R}^N$: per-channel mean log-firing rate
- $\boldsymbol{Q} \in \mathbb{R}^{K \times K}$: latent process noise covariance
- $\boldsymbol{\mu}_0 \in \mathbb{R}^K$, $\boldsymbol{\Sigma}_0 \in \mathbb{R}^{K \times K}$: initial state mean and covariance
- $\boldsymbol{\theta} = \{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{b}, \boldsymbol{C}, \boldsymbol{d}, \boldsymbol{Q}, \boldsymbol{\mu_0}, \boldsymbol{\Sigma_0}\}$: full parameter set

$$V_{ij} = \frac{1}{Z(w)} \exp\left(-\frac{(i-j)^2}{2w^2}\right) \quad Z(w) = \sum_{k=-\infty}^{\infty} \exp\left(-\frac{k^2}{2w^2}\right)$$

Overview

1. **Control trials**: Monkey performs control reaches to the target
2. We fit the PLDS to the control reaches to obtain $\hat{\boldsymbol{A}}_{\text{ctrl}}, \hat{\boldsymbol{C}}_{\text{ctrl}}, \hat{\boldsymbol{d}}_{\text{ctrl}}, \hat{\boldsymbol{Q}}_{\text{ctrl}}$
3. **Stim trials**: During each trial, perform online filtering using the fitted parameters $\hat{\boldsymbol{\theta}}_{\text{i}-1}$, to get $\{\boldsymbol{x}_{i,1:t}\}$ in real time. This gives us the state at which the stimulation happens.
   (a) At the first trial, we initialize with $\hat{\boldsymbol{\theta}}_{\text{i}-1} = \{\hat{\boldsymbol{A}}_{\text{ctrl}}, \hat{\boldsymbol{B}}_{0|0}, \hat{\boldsymbol{b}}_{0|0}, \hat{\boldsymbol{C}}_{\text{ctrl}}, \hat{\boldsymbol{d}}_{\text{ctrl}}, \hat{\boldsymbol{Q}}_{\text{ctrl}}, \hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\Sigma}}_0\}$, where $\hat{\boldsymbol{B}}_{0|0} = \boldsymbol{0}$, $\hat{\boldsymbol{\Sigma}}_{B_{0|0}} = \sigma_B^2 \boldsymbol{I}$, and $\hat{\boldsymbol{b}}_{0|0} = \boldsymbol{0}$.
   (b) This means that we have no information about the stim-to-latent mapping or the perturbation threshold and the first stimulation sites are chosen at random.
4. **Model fitting**: Between each trial, we fit the spike counts $\{\boldsymbol{y}_{i,1:T}\}$ to the model using the Extended Kalman Filter (EKF)-style integral and stochastic gradient descent (SGD). This gives us updated parameters $\hat{\boldsymbol{\theta}}_{\text{i}} \leftarrow \hat{\boldsymbol{\theta}}_{\text{i}-1}$
   (a) The previous trial would "warm-start" the convergence of the next trial, so we only need to run a few iterations of EKF and SGD between each trial to get a converging estimate of the parameters.
5. **Acquisition:**
   (a) **Recursive least squares**: Using RLS to track $\hat{\boldsymbol{B}}_i$ and $\hat{\boldsymbol{\Sigma}}_{B_{i|i}}$ gives us an estimate of the covariance in the stim-to-latent mapping. This step uses $\hat{\boldsymbol{A}}_i, \hat{\boldsymbol{Q}}_i, \hat{\boldsymbol{x}}_{i,t|T}, \boldsymbol{P}_{i,t-1|T}$, and $\hat{\boldsymbol{d}}_i$ estimated from steps 4.
   (b) **Stim selection**: Choose the next stimulation $\boldsymbol{s}_{i+1}$ using the covariance estimate of $\hat{\boldsymbol{B}}_{i+1|i}$ ($\hat{\boldsymbol{\Sigma}}_{B_{i+1|i}}$), which is the uncertainty in the stim-to-latent mapping after trial $i+1$ after a given stimulation.
6. Perform the next stimulation and repeat 3-5.

Model fitting

State-space equation (PLDS)

$$
\begin{aligned}
\boldsymbol{x}_0 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
\boldsymbol{u} &= \text{softplus}\big(\boldsymbol{V}\boldsymbol{s}_{t_{stim}} + \boldsymbol{b}\big) \\
\boldsymbol{x}_t &= \boldsymbol{A}\boldsymbol{x}_{t-1} + \boldsymbol{B}\boldsymbol{u} + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \\
\boldsymbol{y}_t &\sim \text{Poisson}\big(\text{softplus}(\boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{d})\big)
\end{aligned}
$$

We denote two distinct indices throughout: Ex: for $\boldsymbol{A}_{i,t}$

- $i = 1, 2, \ldots$ indexes *trials*
- $t = 1, \ldots, T$ indexes *within-trial timesteps*

Initialization (priors at first stim trial)

We first collect a calibration session with the monkey performing control reaches, which identifies the initial model parameters (and spike detection templates).

Control trial initialization ($i = 0$)

Initialize and fit PLDS on control trials:

$$
\begin{aligned}
\boldsymbol{A}_{0,\text{ctrl}} &= \boldsymbol{I}_K \\
\boldsymbol{C}_{0,\text{ctrl}} &\sim \mathcal{N}(\boldsymbol{0},\, \sigma_C^2\, \boldsymbol{I}_{N\times K}) \\
\boldsymbol{d}_{0,\text{ctrl}} &= \log \bar{\boldsymbol{y}}_{0,\text{ctrl}} \\
\boldsymbol{\mu}_{0,\text{ctrl}} &= \boldsymbol{0}_K \\
\boldsymbol{\Sigma}_{0,\text{ctrl}} &= \boldsymbol{I}_K
\end{aligned}
$$

$\Downarrow$ fit on control data

$$
\hat{\boldsymbol{A}}_{\text{ctrl}}, \hat{\boldsymbol{C}}_{\text{ctrl}}, \hat{\boldsymbol{d}}_{\text{ctrl}}, \hat{\boldsymbol{Q}}_{\text{ctrl}}
$$

First stim trial initialization ($i = 1$)

Warm start with fitted control parameters:

$$
\begin{aligned}
\hat{\boldsymbol{A}}_1 &= \hat{\boldsymbol{A}}_{\text{ctrl}} \\
\hat{\boldsymbol{C}}_1 &= \hat{\boldsymbol{C}}_{\text{ctrl}} \\
\hat{\boldsymbol{d}}_1 &= \log \bar{\boldsymbol{y}}_1
\end{aligned}
$$

Stim parameters are initialized via:

$$
\begin{aligned}
\hat{\boldsymbol{B}}_{0|0} &= \boldsymbol{0}_{K\times N} \\
\hat{\boldsymbol{\Sigma}}_{B_{0|0}} &= \sigma_B^2\, \boldsymbol{I} \\
\hat{\boldsymbol{b}}_{0|0} &= \boldsymbol{0}_K
\end{aligned}
$$

where:

- σ_B^2: controls prior uncertainty over the stim-to-latent mapping
- σ_C^2: controls prior uncertainty over the latent-to-observation mapping
- $\bar{\boldsymbol{y}}_{\text{ctrl}}$: mean spike count from control trials
- $\hat{\boldsymbol{\Sigma}}_{B_{0|0}} = \sigma_B^2\, \boldsymbol{I}$ (spherical prior) means $\boldsymbol{s}_1$ is chosen at random

Stim trial initialization ($i \geq 2$) uses the previous trial's fitted parameters $\hat{\boldsymbol{\theta}}_{i-1}$ as the prior for the next trial.

Online forward filtering during a trial

During a trial, we estimate the latent state $\boldsymbol{x}_{i,t}$ in real time using the EKF forward pass with the current parameter estimates $\hat{\boldsymbol{\theta}}_{i-1}$. This gives us an estimate of the latent state in real time.

EKF forward pass

The marginal likelihood of the latent trajectory is:

$$\log p(\boldsymbol{y}_{1:T}|\boldsymbol{\theta}) = \log \int p(\boldsymbol{y}_{1:T} \mid \boldsymbol{x}_{1:T}; \boldsymbol{\theta})\, p(\boldsymbol{x}_{1:T}; \boldsymbol{\theta})\, d\boldsymbol{x}_{1:T}$$

$$\boldsymbol{y}_t \sim \text{Poisson}\big(\text{softplus}(\boldsymbol{C}\boldsymbol{x}_t + \boldsymbol{d})\big)$$

$$\boldsymbol{\theta} = \{\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{b}, \boldsymbol{C}, \boldsymbol{d}, \boldsymbol{Q}, \boldsymbol{\mu_0}, \boldsymbol{\Sigma_0}\}$$

Because the Poisson emission $p(\boldsymbol{y}_{1:t} \mid \boldsymbol{x}_{1:t}; \boldsymbol{\theta})$ is non-Gaussian, the integral is intractable. At time $t-1$, we need to approximate it by linearizing the emission around $\hat{\boldsymbol{x}}_{t|t-1}$, the current filtered mean (latent state), yielding a Gaussian approximation to the predictive distribution

$$p(\boldsymbol{y}_t \mid \boldsymbol{y}_{1:t-1}; \boldsymbol{\theta}) \approx \mathcal{N}(\boldsymbol{\lambda}_{t|t-1},\, \boldsymbol{S}_t), \qquad \boldsymbol{\lambda}_{t|t-1} = \text{softplus}\Big(\boldsymbol{C}\hat{\boldsymbol{x}}_{t|t-1} + \boldsymbol{d}\Big)$$

where $\boldsymbol{S}_t$ is the estimated covariance.

$$\mathcal{L}(\boldsymbol{\theta}) \approx \sum_{t=1}^{T} \log \mathcal{N}(\boldsymbol{y}_t |\, \boldsymbol{\lambda}_{t|t-1},\, \boldsymbol{S}_t)$$

Emission Jacobian

In the EKF formulation, we approximate the observation operator $\boldsymbol{H}_t$ with the Emission Jacobian described here: For $\boldsymbol{\lambda}_t = \text{softplus}(\boldsymbol{C}\hat{\boldsymbol{x}}_t + \boldsymbol{d})$, we have the Jacobian (log-rate with respect to the latent state) is:

$$\boldsymbol{H}_t = \left.\frac{\partial \boldsymbol{\lambda}_t}{\partial \boldsymbol{x}_t}\right|_{\hat{\boldsymbol{x}}_t} = \text{diag}(\sigma(\boldsymbol{C}\hat{\boldsymbol{x}}_t + \boldsymbol{d}))\, \boldsymbol{C} \in \mathbb{R}^{N \times K}$$

where $\sigma(\cdot)$ is the sigmoid function (derivative of softplus). We calculate the Emission Jacobian $\boldsymbol{H}_t$ to linearize the non-Gaussian Poisson emission and plug into the Kalman equations. This propagates the Gaussian estimations of latent states $\boldsymbol{x}_t$ one time step forward each, which we can plug in the Kalman equations.

EKF forward pass prediction update

$$\begin{aligned}
\hat{\boldsymbol{x}}_{t|t-1} &= \boldsymbol{A}\hat{\boldsymbol{x}}_{t-1|t-1} + \boldsymbol{B}\,\text{softplus}(\boldsymbol{V}\boldsymbol{s}_t + \boldsymbol{b}) \\
\boldsymbol{P}_{t|t-1} &= \boldsymbol{A}\boldsymbol{P}_{t-1|t-1}\boldsymbol{A}^\top + \boldsymbol{Q} \\
\boldsymbol{S}_t &= \boldsymbol{H}_t\boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^\top + \text{diag}(\boldsymbol{\lambda}_{t|t-1}) \\
\boldsymbol{K}_t &= \boldsymbol{P}_{t|t-1}\boldsymbol{H}_t^\top \boldsymbol{S}_t^{-1} \\
\hat{\boldsymbol{x}}_{t|t} &= \hat{\boldsymbol{x}}_{t|t-1} + \boldsymbol{K}_t(\boldsymbol{y}_t - \boldsymbol{\lambda}_{t|t-1}) \\
\boldsymbol{P}_{t|t} &= (\boldsymbol{I} - \boldsymbol{K}_t\boldsymbol{H}_t)\boldsymbol{P}_{t|t-1}
\end{aligned}$$

- $\hat{\boldsymbol{x}}_{t|t-1} \in \mathbb{R}^K$: predicted latent state mean
- $\hat{\boldsymbol{x}}_{t|t} \in \mathbb{R}^K$: filtered latent state mean
- $\boldsymbol{P}_{t|t-1} \in \mathbb{R}^{K \times K}$: predicted latent state covariance
- $\boldsymbol{P}_{t|t} \in \mathbb{R}^{K \times K}$: filtered latent state covariance
- $\boldsymbol{S}_t \in \mathbb{R}^{N \times N}$: innovation covariance
- $\boldsymbol{K}_t \in \mathbb{R}^{K \times N}$: Kalman gain
- $(\boldsymbol{y}_t - \boldsymbol{\lambda}_{t|t-1})$: innovation (residual)
- $\boldsymbol{Q} \in \mathbb{R}^{K \times K}$: latent process noise covariance
- $\text{diag}(\boldsymbol{\lambda}_{t|t-1})$: Poisson observation noise
- $\boldsymbol{H}_t \in \mathbb{R}^{N \times K}$: locally linear approximation of emission matrix

Offline parameter update (Outline)

At the end of each trial i, the following steps are performed in order:

1. "E step" **EKF forward pass + RTS smoother**: Run the EKF forward over $\boldsymbol{y}_{i,1:T}$ with $\hat{\boldsymbol{\theta}}_{i-1}$ fixed, then run the RTS smoother backward to get filtered means and covariances $\{\hat{\boldsymbol{x}}_{i,1:T}, \boldsymbol{P}_{i,1:T}\}$. $\{\hat{\boldsymbol{x}}_{i,t|t}, \boldsymbol{P}_{i,t|t}\}_{t=1}^{T}$ is the lag-one cross-covariance needed for the M-steps below.

2. "M step" **Factorized estimation of parameters**: We factorize the estimation of parameters to get point estimates of parameters with no closed form ($\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$, $\hat{\boldsymbol{b}}_i$), but still getting covariance estimates for those that we can using EM. This also keeps the faster convergence rate of the EM algorithm.

   (a) **SGD update $\hat{\boldsymbol{\theta}}_i$** Take one gradient step on the single-trial EKF marginal log-likelihood $\mathcal{L}_i = \sum_{t=1}^{T} \log \mathcal{N}(\boldsymbol{y}_{i,t} \mid \boldsymbol{\lambda}_{i,t|t-1}, \boldsymbol{S}_{i,t})$:

$$\hat{\boldsymbol{\theta}}_i \leftarrow \hat{\boldsymbol{\theta}}_{i-1} - \eta_i \nabla_{\boldsymbol{C},\boldsymbol{d}} \left[-\mathcal{L}_i(\hat{\boldsymbol{\theta}}_{i-1})\right]$$

3. **Repeat 1 and 2 until convergence**:

   (a) Steps (a)-(c) are handled in the DYNAMAX package using conditional moments gaussian smoother (CMGS), but we lay out the details below.

   (b) We may need hyperparameter optimization on the batch size and learning rate.

   (c) This process can terminate early if the next trial happens before convergence.

   (d) We will need to analyze our **convergence criterion** and if we need an algorithm that is quicker / more accurate. (Low rank prior of loading matrix, instead of a state-space formulation)

EKF forward pass + RTS smoother

The EKF forward pass is identical to the online version. (see **Online forward filtering**)

RTS smoother backward pass

$$
\begin{aligned}
\boldsymbol{G}_t &= \boldsymbol{P}_{t|t}\boldsymbol{A}^\top \boldsymbol{P}_{t+1|t}^{-1} \\
\hat{\boldsymbol{x}}_{t|T} &= \hat{\boldsymbol{x}}_{t|t} + \boldsymbol{G}_t\Big(\hat{\boldsymbol{x}}_{t+1|T} - \hat{\boldsymbol{x}}_{t+1|t}\Big) \\
\boldsymbol{P}_{t|T} &= \boldsymbol{P}_{t|t} + \boldsymbol{G}_t\Big(\boldsymbol{P}_{t+1|T} - \boldsymbol{P}_{t+1|t}\Big)\,\boldsymbol{G}_t^\top
\end{aligned}
$$

initialized at $\hat{\boldsymbol{x}}_{T|T}$, $\boldsymbol{P}_{T|T}$ and run backward for $t = T-1, \ldots, 1$.

SGD parameter update $\hat{\boldsymbol{\theta}}_i$

Parameters related to the Poisson observation equation ($\hat{\boldsymbol{C}}_i$, $\hat{\boldsymbol{d}}_i$) and the stim-to-latent threshold ($\hat{\boldsymbol{b}}_i$) have no closed form. So, we made a design choice to update the parameter set by using SGD and autodiff through the posterior:

$$
\boldsymbol{\theta}^{(\tau+1)} = \boldsymbol{\theta}^{(\tau)} - \eta_\tau \, \nabla_{\boldsymbol{\theta}} \left[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\right]
$$

where η_τ is the effective learning rate defined by the Adam optimizer. More specifically:

$$
\begin{aligned}
\boldsymbol{m}_\tau &= \beta_1 \boldsymbol{m}_{\tau-1} + (1-\beta_1)\,\nabla_{\boldsymbol{\theta}}\big[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\big] && \text{(momentum)} \\
\boldsymbol{v}_\tau &= \beta_2 \boldsymbol{v}_{\tau-1} + (1-\beta_2)\,\nabla_{\boldsymbol{\theta}}\big[-\mathcal{L}(\boldsymbol{\theta}^{(\tau)})\big]^2 && \text{(RMSprop)} \\
\hat{\boldsymbol{m}}_\tau &= \frac{\boldsymbol{m}_\tau}{1-\beta_1^\tau}, \qquad \hat{\boldsymbol{v}}_\tau = \frac{\boldsymbol{v}_\tau}{1-\beta_2^\tau} && \text{(bias correction)} \\
\boldsymbol{\theta}^{(\tau+1)} &= \boldsymbol{\theta}^{(\tau)} - \frac{\alpha}{\sqrt{\hat{\boldsymbol{v}}_\tau} + \epsilon}\,\hat{\boldsymbol{m}}_\tau && \text{(parameter update)}
\end{aligned}
$$

where $\boldsymbol{m}_\tau$ and $\boldsymbol{v}_\tau$ are the first and second moment estimates, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ are the default hyperparameters, and α is the base learning rate. Gradients are computed via automatic differentiation in JAX in the DYNAMAX package. We get $\hat{\boldsymbol{\theta}}_i$ after convergence.

Getting the posterior $p(\hat{\boldsymbol{B}}_{i|i-1} \mid \boldsymbol{y}_{1:i-1}, \boldsymbol{s}_i)$

Stimulation pattern $\boldsymbol{s}_i \in \{0,1\}^N$ is fixed for trial i and happens at t_{stim}.

$$\begin{aligned}
\boldsymbol{x}_{i,0} &\sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) \\
\boldsymbol{x}_{i,t} &= \boldsymbol{A}\boldsymbol{x}_{i,t-1} + \boldsymbol{B}\boldsymbol{u}_i + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}) \\
\boldsymbol{y}_{i,t} &\sim \text{Poisson}\Big(\text{softplus}\big(\boldsymbol{C}\boldsymbol{x}_{i,t} + \boldsymbol{d}\big)\Big)
\end{aligned}$$

We assume we can factorize across the $\boldsymbol{\theta}$s (Much like coordinate ascent variational inference). We get the estimates of $\boldsymbol{\theta} \setminus \boldsymbol{B}$ known from PLDS fit via EKF SGD from trials $1, \ldots, i-1$ and treated as fixed at trial i. $\boldsymbol{s}_t$ is known:

$$\boldsymbol{\theta} \setminus \boldsymbol{B} = \{\boldsymbol{A},\, \boldsymbol{C},\, \boldsymbol{d},\, \boldsymbol{Q},\, \boldsymbol{\mu}_0,\, \boldsymbol{\Sigma}_0\} = \hat{\boldsymbol{\theta}}_{i-1} \setminus \boldsymbol{B}$$

$$\begin{aligned}
p(\boldsymbol{\theta} \mid \boldsymbol{y}_{1:t}) &\propto p(\boldsymbol{y}_{1:t} \mid \boldsymbol{\theta}) \cdot p(\boldsymbol{\theta}) \\
p(\boldsymbol{B} \mid \boldsymbol{y}_{1:t}, \boldsymbol{s}_t) &\propto p(\boldsymbol{y}_{1:t} \mid \boldsymbol{B},\, \boldsymbol{\theta} \setminus \boldsymbol{B}) \cdot p(\boldsymbol{B}) \cdot p(\boldsymbol{A}) \cdot p(\boldsymbol{C}) \cdots \quad \text{(by independence of } \boldsymbol{\theta} \text{ components)}
\end{aligned}$$

Absorbing $p(\boldsymbol{\theta} \setminus \boldsymbol{B})$ into the constant and expanding over $\boldsymbol{x}_{1:t}$:

$$p(\boldsymbol{B} \mid \boldsymbol{y}_{1:t}, \boldsymbol{s}_t) \propto \underbrace{p(\boldsymbol{B})}_{\text{Initialize via } \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_0^B)} \int \underbrace{p(\boldsymbol{y}_{1:t} \mid \boldsymbol{x}_{1:t}, \boldsymbol{B}, \boldsymbol{\theta} \setminus \boldsymbol{B})}_{\text{Poisson emission}} \underbrace{p(\boldsymbol{x}_{1:t} \mid \boldsymbol{s}_t, \boldsymbol{\theta})}_{\text{depends on } \boldsymbol{B}, \text{can get from EKF}} d\boldsymbol{x}_{1:t} \tag{1}$$

Likelihood after marginalizing out $\boldsymbol{x}_{1:t}$:

$$p(\boldsymbol{y}_{1:t} \mid \boldsymbol{\theta}) = \int \underbrace{p(\boldsymbol{y}_{1:t} \mid \boldsymbol{x}_{1:t}, \boldsymbol{\theta})}_{\text{Poisson emission, need EKF approx.}} \underbrace{p(\boldsymbol{x}_{1:t} \mid \boldsymbol{s}_t, \boldsymbol{\theta})}_{\text{Gaussian dynamics prior}} d\boldsymbol{x}_{1:t}$$

$$\boldsymbol{\theta} = \{\boldsymbol{A},\, \boldsymbol{B},\, \boldsymbol{C},\, \boldsymbol{d},\, \boldsymbol{Q},\, \boldsymbol{\mu}_0,\, \boldsymbol{\Sigma}_0\}$$

Bayes':

$$\underbrace{p(\boldsymbol{\theta} \mid \boldsymbol{y}_{1:t})}_{\text{posterior}} = \frac{\overbrace{p(\boldsymbol{y}_{1:t} \mid \boldsymbol{\theta})}^{\text{likelihood}} \cdot \overbrace{p(\boldsymbol{\theta})}^{\text{prior}}}{\underbrace{p(\boldsymbol{y}_{1:t})}_{\text{evidence, constant in } \boldsymbol{\theta}}}$$

Acquisition function: recursive Bayesian update with full noise covariance

We need to select the next stimulation pattern by minimizing the determinant of the covariance matrix of $\boldsymbol{B}$. However, we don't have access to it because we opt for SGD to tackle the intractable posterior, which only gives us point estimates. Therefore, we use a recursive least squares (RLS) approach to track a posterior over each row of $\boldsymbol{B}$. We update $\boldsymbol{\Sigma}_{B_{i|i}}$ using noise covariance $\boldsymbol{R}_i \in \mathbb{R}^{K\times K}$ calculated by the EKF/RTS smoothed latents. Then we calculate the covariance $\boldsymbol{\Sigma}_{B_{i+1|i}}$, which we can then use to inform our choice of the next stimulation pattern.

One-step ahead prediction of latents gives us the estimate of the perturbation

Define the effective stimulation input $\boldsymbol{u}_i = \text{softplus}(\boldsymbol{V}\boldsymbol{s}_i + \boldsymbol{b}) \in \mathbb{R}^N$. From the EKF forward pass + RTS smoother at trial i,

$$\boldsymbol{z}_i = \boldsymbol{B}\boldsymbol{u}_i + \boldsymbol{\varepsilon}_i, \qquad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_i),$$

where:

$$\boldsymbol{R}_i = \text{Cov}(\hat{\boldsymbol{x}}_{i,t|T}) + \hat{\boldsymbol{A}}_i \text{Cov}(\hat{\boldsymbol{x}}_{i,t-1|T})\hat{\boldsymbol{A}}_i^\top + \hat{\boldsymbol{Q}}_i \quad \in \mathbb{R}^{K\times K}$$

is the covariance of the stimulation-induced latent perturbation estimate. Here we assume the rows of $\boldsymbol{B}$ (how stim affects each latent) are independent. We are also ignoring $\text{Cov}(\hat{\boldsymbol{A}})$.

This gives us RLS tracking for each row of $\boldsymbol{B}$ (although $\boldsymbol{R}_i$ still contributes to the uncertainty for each row). Let row k carries its own prior $\boldsymbol{w}_k := \boldsymbol{B}_{k,:}^\top \sim \mathcal{N}\big(\hat{\boldsymbol{w}}_{i-1}^{(k)}, \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)}\big) \quad \in \mathbb{R}^{N\times 1}$. Note: full tracking of covariance would require tracking a $NK \times NK$ matrix. We are dropping the cross-latent covariance for now.

Updating posterior (Bayes rule)

Since both the prior and the likelihood are Gaussian, the posterior

$$p(\boldsymbol{B} \mid \boldsymbol{z}_{1:i}) \propto p(\boldsymbol{z}_i \mid \boldsymbol{B})\, p(\boldsymbol{B} \mid \boldsymbol{z}_{1:i-1})$$

is also Gaussian, the standard Bayesian linear-regression / RLS gain formula gives us a balance between the prior and the information gained from each trial.

Predictive covariance of the observation

For row k, the relevant predictive quantity is the variance of the scalar $\boldsymbol{w}_k^\top \boldsymbol{u}_i$ under the current prior $\boldsymbol{w}_k \sim \mathcal{N}(\hat{\boldsymbol{w}}_{i-1}^{(k)}, \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)})$. Expanding the variance directly from its definition,

$$\begin{aligned}
\text{Var}(\boldsymbol{w}_k^\top \boldsymbol{u}_i) &= \mathbb{E}\Big[(\boldsymbol{w}_k^\top \boldsymbol{u}_i)^2\Big] - \big(\mathbb{E}[\boldsymbol{w}_k^\top \boldsymbol{u}_i]\big)^2 \\
&= \mathbb{E}\Big[\boldsymbol{u}_i^\top \boldsymbol{w}_k \boldsymbol{w}_k^\top \boldsymbol{u}_i\Big] - \boldsymbol{u}_i^\top\, \mathbb{E}[\boldsymbol{w}_k]\mathbb{E}[\boldsymbol{w}_k]^\top\, \boldsymbol{u}_i \\
&= \boldsymbol{u}_i^\top \Big(\mathbb{E}\Big[\boldsymbol{w}_k \boldsymbol{w}_k^\top\Big] - \mathbb{E}[\boldsymbol{w}_k]\mathbb{E}[\boldsymbol{w}_k]^\top\Big)\boldsymbol{u}_i \\
&= \boldsymbol{u}_i^\top\, \text{Cov}(\boldsymbol{w}_k)\, \boldsymbol{u}_i.
\end{aligned}$$

Using $\mathbb{E}[\boldsymbol{w}_k] = \hat{\boldsymbol{w}}_{i-1}^{(k)}$ and $\text{Cov}(\boldsymbol{w}_k) = \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)}$ from the prior, this confirms

$$\sigma_i^{(k)} := \text{Var}(\boldsymbol{w}_k^\top \boldsymbol{u}_i) = \boldsymbol{u}_i^\top \hat{\boldsymbol{\Sigma}}_{B_{i-1}}^{(k)} \boldsymbol{u}_i,$$

This means, that row k's contribution to the predictive uncertainty of $\boldsymbol{z}_i$ is its prior covariance projected onto the stimulated direction $\boldsymbol{u}_i$.

Since each row contributes independent prior variance $\sigma_i^{(k)}$ to its own directions. The contribution to the predictive covariance of $\boldsymbol{z}_i$ is the diagonal:

$$\boldsymbol{D}_i := \operatorname{diag}\big(\sigma_i^{(1)}, \ldots, \sigma_i^{(K)}\big).$$

Adding the observation noise from the observed $\boldsymbol{R}_i$ from $\boldsymbol{z}_i = \boldsymbol{B}\boldsymbol{u}_i + \boldsymbol{\varepsilon}_i$ gives the full predictive covariance,

$$\boldsymbol{S}_i := \boldsymbol{D}_i + \boldsymbol{R}_i \in \mathbb{R}^{K \times K},$$

These two components contribute to the predictive covariance of $\boldsymbol{z}_i$ by:

1. $\boldsymbol{D}_i$ is the uncertainty in the prediction of $\boldsymbol{z}_i$ due to the uncertainty in $\boldsymbol{B}$
2. $\boldsymbol{R}_i$ is the uncertainty in the observation of $\boldsymbol{z}_i$ due to the noise in the measurement process, which we get from the noise in the latents. Note: This noise can correlate across latents

$\boldsymbol{S}_i \in \mathbb{R}^{K \times K}$ is the only matrix that needs inverting for precision.

RLS gain and update via joint Gaussian conditioning

Since $\boldsymbol{w}_k$ is Gaussian and $\boldsymbol{z}_i = \boldsymbol{B}\boldsymbol{u}_i + \boldsymbol{\varepsilon}_i$ is a linear function of $\boldsymbol{B}$ (hence of $\boldsymbol{w}_k$) plus independent Gaussian noise, $(\boldsymbol{w}_k, \boldsymbol{z}_i)$ are jointly Gaussian. For jointly Gaussian $(\boldsymbol{X}, \boldsymbol{Y})$, the conditional distribution has the form

$$\boldsymbol{X} \mid \boldsymbol{Y} \;\sim\; \mathcal{N}\Big(\boldsymbol{\mu}_X + \boldsymbol{\Sigma}_{XY}\boldsymbol{\Sigma}_Y^{-1}(\boldsymbol{Y} - \boldsymbol{\mu}_Y),\;\; \boldsymbol{\Sigma}_X - \boldsymbol{\Sigma}_{XY}\boldsymbol{\Sigma}_Y^{-1}\boldsymbol{\Sigma}_{YX}\Big).$$

Here, we are calculating the posterior update for row k where:

1. $\boldsymbol{X} = \boldsymbol{w}_k$
2. $\boldsymbol{Y} = \boldsymbol{z}_i$
3. $\boldsymbol{\mu}_Y = \mathbb{E}[\boldsymbol{z}_i] = \hat{\boldsymbol{B}}_{i-1}\boldsymbol{u}_i$
4. $\boldsymbol{\Sigma}_Y = \boldsymbol{S}_i = \boldsymbol{D}_i + \boldsymbol{R}_i$ is the predictive covariance of $\boldsymbol{z}_i$ derived above.
5. $\boldsymbol{\Sigma}_{XY} =$

Mean update

Covariance update

Choosing the next stimulation pattern $\boldsymbol{u}_{i+1}$ Gaussian case TODO: Update for RLS

For adaptive selection of stimulation patterns, before each trial $i+1$, we want to select $\boldsymbol{u}_{i+1}$ that maximizes the expected information gain about $\boldsymbol{B}$. Equivalently, this would be selecting $\boldsymbol{u}_{i+1}$ that minimizes the determinant of the covariance of $\boldsymbol{B}$ after trial $i+1$ ($\hat{\boldsymbol{\Sigma}}_{B_{i+1|i+1}}$).

Let $\boldsymbol{\Sigma}^{-1}_{B_{i\|i}}$	be the precision matrix of $\boldsymbol{B}$ after trial i
Let $\boldsymbol{\Delta}_{i+1} = \boldsymbol{B}\boldsymbol{u}_{i+1}$	be the latent perturbation induced by stimulation pattern $\boldsymbol{u}_{i+1}$
Let β	be how we scale our data relative to our prior (could be $1/n$)

In terms of precision, we can formulate the Gaussian case as:

$$\boldsymbol{\Sigma}^{-1}_{B_{i+1|i}} = \boldsymbol{\Sigma}^{-1}_{B_{i|i}} + \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]\mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]^\top$$

Taking the determinant and applying the matrix determinant lemma:

$$\begin{aligned}
\det \boldsymbol{\Sigma}^{-1}_{B_{i+1|i}} &= \det\Big(\boldsymbol{\Sigma}^{-1}_{B_{i|i}} + \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]\mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]^\top\Big) \\
&= \Big(1 + \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]^\top \boldsymbol{\Sigma}^B_i\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]\Big) \det\big(\boldsymbol{\Sigma}^{-1}_{B_{i|i}}\big)
\end{aligned}$$

Since we minimize over $\boldsymbol{u}_{i+1}$, the objective function is therefore:

$$\begin{aligned}
\arg\max_{\boldsymbol{u}}\ \det\big(\boldsymbol{\Sigma}^{-1}_{B_{i+1|i}}\big) &= \arg\max_{\boldsymbol{u}}\ \Big(1 + \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]^\top \boldsymbol{\Sigma}^B_i\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]\Big) \det\big(\boldsymbol{\Sigma}^{-1}_{B_{i|i}}\big) \\
&= \arg\max_{\boldsymbol{u}}\ \beta\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]^\top \boldsymbol{\Sigma}^B_i\, \mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}]
\end{aligned}$$

We can further expand out the expected perturbation in latent space:

$$\mathbb{E}[\boldsymbol{\Delta}_{i+1} \mid \boldsymbol{u}_{i+1}] = \hat{\boldsymbol{B}}_i \boldsymbol{u}_{i+1}$$

where $\hat{\boldsymbol{B}}_i$ is estimated from trials $1, \ldots, i$, giving the final objective:

$$\arg\max_{\boldsymbol{u}}\ \beta\, \boldsymbol{u}_{i+1}^\top \hat{\boldsymbol{B}}_i^\top \boldsymbol{\Sigma}^B_i\, \hat{\boldsymbol{B}}_i \boldsymbol{u}_{i+1}$$

TODO

1. We will need a **compositional model** of some sort to handle the fact that if there is behavioral change, there is likely a change in the latent dynamics as well, which will affect the estimation of $\boldsymbol{B}$. Change point detection? Or construct using known time-scale of behavioral / neural feedback
2. We will need to analyze our **convergence criterion** and if we need an algorithm that is quicker / more accurate. (Low rank prior of loading matrix, instead of a state-space formulation):
3. Might need a **greedy search** for the acquisition function to not focus on the same noisy channels repeatedly.
4. This is for one timepoint, don't worry too much about some kind of t to t+k information update, before deciding on the next step, where t+k is the next stim LINOCS?
5. Do we need to expand to NKxNK for the precision matrix? instead of a K X N x N matrix.
6. Tried fitting the PLDS with a step term ($\boldsymbol{C} \times \text{step}(t)$) before, but doesn't seem to do well.
7. Try to find closed form after plugging in $\mathbb{E}[\boldsymbol{B}]$?

- Since s is Bernoulli (discrete), projected or coordinate or proximal descent to solve for $\boldsymbol{u}$
- Projected is projecting the final to discrete, proximal is projecting each step